Midterm

Active Directory Domain Services (ADDS)

- Definition: What is ADDS?
- Why do we need domain services in a networked environment?

ADDS Database Partitions

- Overview of Schema, Configuration, Domain, and Application partitions.

ADDS Physical and Logical Components

- Physical components
- Sites, replication topology.
- Logical components: Domains, trees, forests, organizational units (OUs).

FSMO Roles (Flexible Single Master Operations)

- Definition, types of FSMO roles, their functions, and how they are distributed.

Global Catalog (GC)

- Role of the Global Catalog in facilitating searches and logins across domains.

IGDLA Model

- Identity → Global Group → Domain Local Group → Access
- Used for structured and scalable permission assignment.

Hyper-V and Hypervisors

- What is Hyper-V?
- Hypervisor types: Type 1 (bare-metal) vs. Type 2 (hosted).
- Use cases and advantages of each.

Hyper-V Virtual Switches

- Types of virtual switches:
    - Internal
    - External
    - Private
- Differences and appropriate use cases.

Checkpoints in Hyper-V

- Definition and usage of checkpoints (snapshots) in virtual machine management.

iSCSI Storage Server and Storage Pool

- Understanding iSCSI protocol and its role in enterprise storage.
- Storage pools and their benefits in virtualization environments.

RAID (Redundant Array of Independent Disks)

- Overview of RAID levels (0, 1, 5, 10).
- Benefits, drawbacks, and required number of disks for each level.

Enterprise Storage Solutions

- Comparison of DAS (Direct-Attached Storage), NAS (Network-Attached Storage), and SAN (Storage Area Network).
- Use cases, advantages, and differences between them.

Storage Layout and Disk Requirements

- Planning storage configurations.
- Minimum disk requirements for different RAID levels and enterprise needs.

Security Identifier (SID)

- What is a SID?
- Why is it important for security and identity management in Windows environments.